An Exact Fold Condition for Mass-Balanced Models of Protein Quality Control

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Abstract

Supplementary material accompanying the paper above. Every number in the caption is recomputed by the script named in it and asserted by the accompanying test suite.

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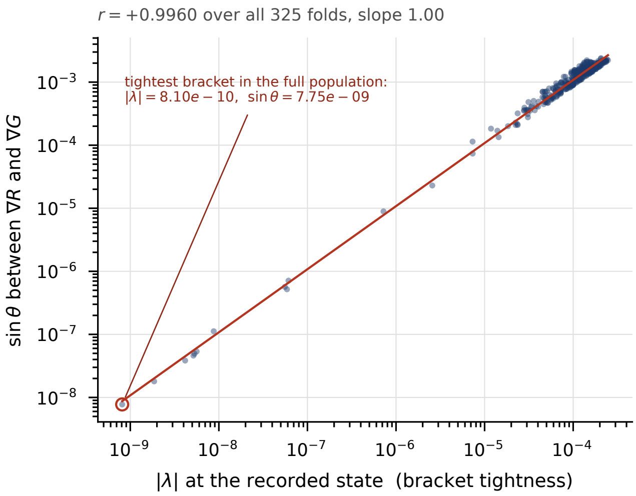

Fig. S1 Parallelism residual against leading eigenvalue, over all 325 load-grid folds with no subsample. The correlation is +0.9960 in log-log with slope 1.000, so $\sin(angle)$ is proportional to bracket looseness rather than merely increasing with it; the tightest-bracketed state has $|\lambda| = 8.10{\times}10^{-10}$ and $\sin(angle) = 7.75{\times}10^{-9}$. The figure also carries the normalisation contrast: normalising by $\max(|\det J|, |\,\mathrm{cross}\,|)$ is 0/0 at an exact fold, and that residual degrades as the bracket tightens (median $3.133{\times}10^{-7}$ on the tighter half against $1.455{\times}10^{-7}$ on the looser, 2.15× worse where it matters most, log-log correlation −0.835, maximum $1.54{\times}10^{-2}$), whereas the gradient normalisation used throughout correlates at +0.041 with a maximum of $1.29{\times}10^{-9}$.

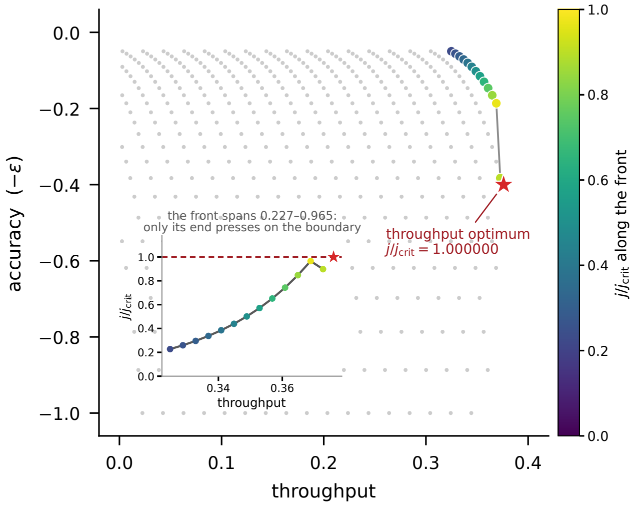

Fig. S2 Strategy space cut by the fold condition: 469 feasible strategies, 13 non-dominated, with j/j_{crit} running 0.2271 to 0.9652 along the front (median 0.5010). The exact throughput optimum sits at $j/j_{crit} = 1.000000$ exactly, on the feasibility boundary; the best grid strategy reaches 0.897488, the gap being discretisation of the grid rather than an interior optimum. Supplementary Section S1 develops a strategy-space consequence of the fold condition: cutting a two-coordinate trade-off surface by the constraint $j(\alpha, R) < j_{crit}(R)$ places the throughput optimum exactly on the feasibility boundary, while along the non-dominated front j/j_{crit} runs 0.227 to 0.965.